An Introduction to Bayesian Methods in Neuropsychology for Psychometrists

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1 Introduction

1.1 About Me

- **Education**
  - B.S. in Psychology (Neuroscience Concentration) from Calvin University
  - M.S. in Quantitative Psychology from Ball State University
- **Current Positions**
  - Lead Psychometrist at Trinity Health West Michigan Medical Group - Neuropsychology Department
  - Adjunct Professor of Psychology at Grand Valley State University
  - Adjunct Instructor of Educational Psychology at Ball State University

1.2 Disclosures & Disclaimers

- I have no financial disclosures or conflicts-of-interests related to this presentation
- This presentation has not been drafted, ideated on, or edited with use of AI or LLMs. Any mistakes made are human error.
- CE credits for viewing this presentation will be provided after the conclusion of the conference and completion of the post-conference survey

1.3 Acknowledgments

- To the National Association of Psychometrists (NAP) Conference Planning Committee and all organizers of this conference
- To my colleagues, Amelia VanderLeest and Blair Shaw
- To the other presenters, before and after me
- To you, my audience

1.4 Learning Objectives

Learning objectives below have been approved by NAP and the Board of Certified Psychometrists (BCP). These objectives will be met with an emphasis on relevancy to the discipline of psychometry. These objectives are written for an audience of psychometrist with at least an undergraduate degree containing at least one course on statistics and/or quantitative methods.

- Learn the general, basic differences between the frequentist and Bayesian frameworks for statistics
- Review the historical and modern prevalent use of frequentist statistics in neuropsychology and cognitive testing
- Appreciate the various developing applications of Bayesian statistics to neuropsychological evaluation - Consider the practical ramifications in test se-

lection and scoring that come with a Bayesian perspective, along with the possible future directions for this approach
- Understand the limitations and ramifications of implementing these methods in a clinic

1.5 My Pitch

- Empirical and evidence-driven nurse analogy
  - We owe it to our patients and clinics to be well-versed in empirical knowledge, even if not used daily
- Improving our stature, variety, skills, and critical thinking
  - With an ever-evolving landscape involving digital testing, AI, and new technology, we must be mindful of how we can train ourselves up in new methods, many of which have roots in Bayesian statistics
- Becoming skilled in statistics may offer an important avenue to future research and educational opportunities!

2 Illustrative Examples

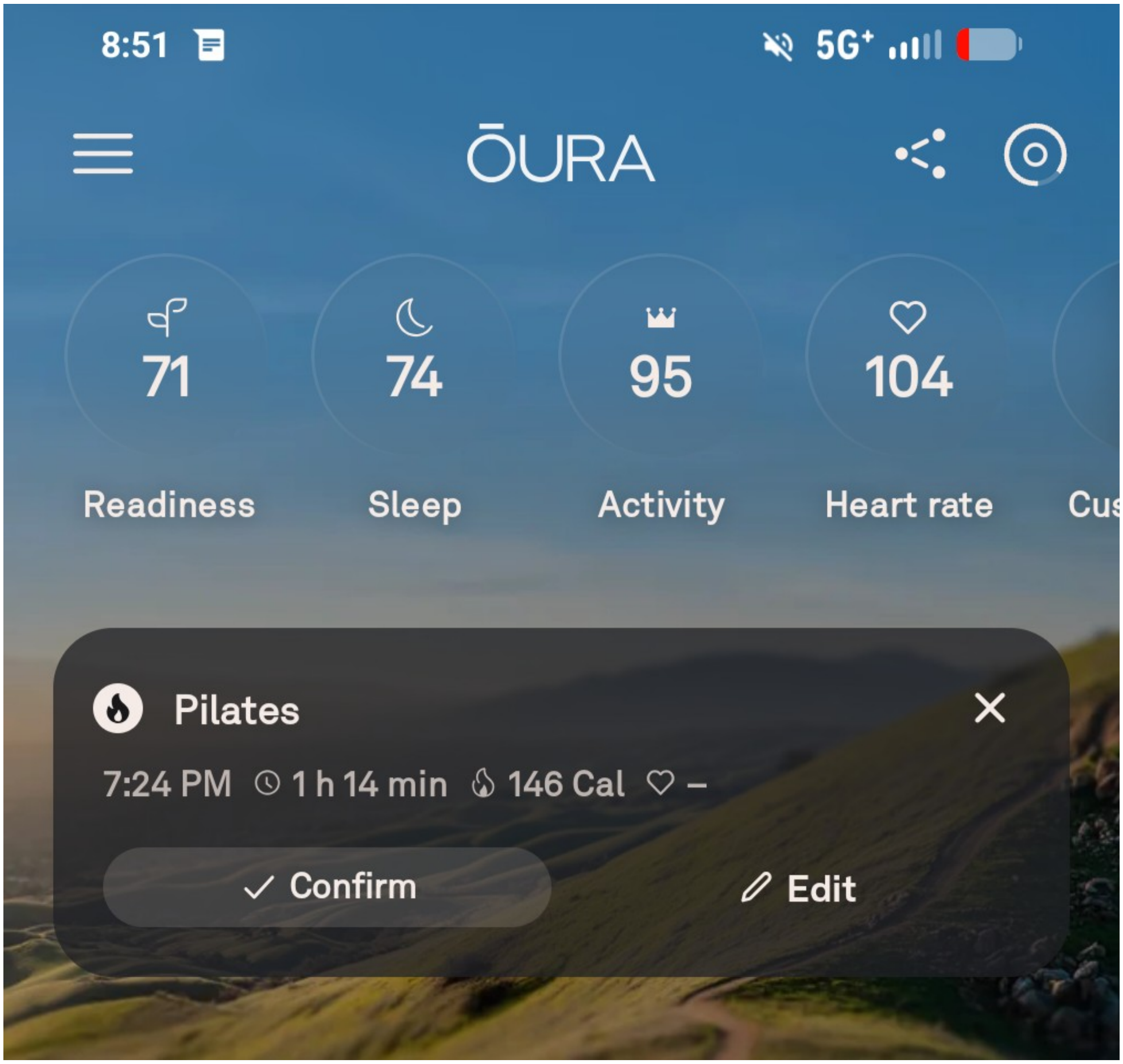
2.1 A Funky Plane Ride

- Patrick is scheduled for a plane ride to go to his favorite conference in the humid city of Baltimore. But he's a nervous flyer…
- Patrick overthinks, and begins to wonder if he is at risk of a plane crash due to weather conditions.
  - 1 in 816,545,929 or 0.000001% chance of dying in a plane crash (The odds of winning the Powerball Jackpot are 1 in 292,200,000)
  - In the following scenario, raise your hand at which you think a plane crash is more than 1% likely to happen

1. Patrick hears there has been a last minute change in flight captain, to a Capt. Tentacles
2. Patrick hears that a flight at the gate next to him has been prevented from landing due to poor weather conditions obstructing landing
3. Patrick hears from his friend, Sandy, that severe thunderstorms are forming in the path of his travel
4. While boarding Patrick notices a sizable teams of mechanics and flight staff looking up into the lower part of the plane, pointing, and arguing
5. While in flight, the plane begins to not only experience major bumps and drops, but also severe crosswinds
6. As he grips his armrests, the person next to him, Sheldon (who is a flight veteran of 15 years) says this is the worst turbulence he's ever experienced

- What is the chance that, after all of this, Patrick's plane crashes?

  - The thought process you employed in this exercise follows the logic of the Bayesian model



2.2 A Questionable Patient

> **Important**
>
> For this example, we are going to play armchair psychologist - do not speculate this in regular practice.

- Fran is a woman concerned for her memory, presenting to the neuropsychology clinic for assessment. She is worried that she may be developing dementia.
- When she comes in, her neuropsychologist, Dr. Sheffield considers the possi-

bilities and explains why they have to do some tests to get answers
  - Based upon demographics and an informed guess Dr. Sheffield initially believes there is a 10% chance of dementia in this patient. She hands off to his psychometrist, Bryton, for cognitive testing
  - In the following scenario, raise your hand at which you think there is a greater than 30% chance that Fran will be diagnosed with dementia

1. Fran misses 1 point on orientation questions, she's off by one on the date, and says she doesn't pay attention because she is retired
2. Bryton notices that he has to repeat himself often and asks if Fran wears hearing aids, she says no, her hearing is fine, he just needs to be more clear when he speaks
3. Bryton records a score on the HVLT (a word memory test) that put's Fran's delayed recall at 0% retention, and a exceptionally low standardized free recall score.
4. Bryton then administers TFLS (an ADLs/functional living test) and Fran, once again, score exceptionally low in all domains. She becomes frustrated, asking why she has to do these.
5. Bryton has to switch out with Grace, his coworker, as he doesn't know how to give one of the tests Dr. Sheffield has assigned. Grace administers a WMS-5 Logical Memory (a story memory test) and, once again, records an exceptionally low standardized recall score.

- If you didn't raise your hand, what other explanations might explain Fran's condition?
  - Again, with weighing this evidence, you employ a mindset in line with the Bayesian model

3 Frequentist vs. Bayesian

3.1 Review of Frequentist Statistics

> **Important**
>
> We are going to get deep in the theory here for just a moment, bear with me! We'll skip the truly theoretical and philosophical stuff - but this will feel the furthest from the work of psychometry.

- Frequentist statistics is a framework of statistics build on the belief that probability is defined as long-run frequency, or relative frequency over an infinite number of trials
  - Simple example: if I were to flip a fair coin an infinite number of times, it would land on head 50% of the time
  - More complex example: if I were to gather an infinite number of samples

from a specified population of US people, the relativity frequency of them having epilepsy would be 1.1% (Kobau et al., 2023)

- Connection to statistics (as we are used to seeing)
  - $p = 0.02$ (p-value) in a comparison between group A and group B on scores on a standardized test
  - Frequentist interpretation: under the assumption that the null hypothesis is true, there is a 2% chance of this difference having randomly occurred in an infinite number of comparisons from these two populations
  - Practical conclusion: if $\alpha = 0.05$, then we can provide evidence that the null hypothesis that these two populations have the same test scores
- Whether you remember it or not, this was almost certainly the way you learned statistics the first time!

3.2 Introduction to Bayesian Statistics

- Bayesian statistics is a framework of statistics built up the belief that probability is the degree of belief of an event
  - Simple example: Initially, I think that the dice are fair, and there is a 1/6 chance of rolling a 6, however, after learning the dice are rigged, I believe that the chance of rolling a 6 are 1/36.
  - Complex example: Upon arriving at the clinic, I believe a person who is 71+ years old has a 13.9% chance of being diagnosed with dementia due to base rates (Plassman et al., 2007). However, after passing several functional and memory measures I believe they only have a 4% chance of being diagnosed with dementia
- Bayesian statistics are rooted in use of Bayes Theorem, that describes the process of updating the degree of belief based upon new information. It is:

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

- Where:
  - A and B are both events; B in particular can be thought of as "new evidence"
  - $P(A)$ is the probability of event A occurring; also called the prior probability
  - $P(B)$ is the probability of event B occurring
  - $P(A|B)$ is the probability of A, given B is true; also called the posterior probability
  - $P(B|A)$ is the probability of A, given B is true; also called the likelihood function
- Effectively, Bayes Theorem is all about how we update our degree of belief in A, given evidence B. This may be a brand new way to think about probabilities for you!

- From the Illustrative Examples earlier:
  - Prior or $P(A)$ of 10% of dementia
  - Additional pieces of evidence modifying that likelihood are B
  - The "hand raise" benchmark of 30% was a description o $P(A|B)$, also know as the posterior probability

> **Important**
>
> Remember, this is a purposefully reductionist example, actual Bayesian results are not fixed percentages, they are distributions of different probabilities!

3.3 How They Differ

- Frequentist statistics are rooted in describing long-run frequency and probabilities of certain events occurring by considering an infinite number of trials
- Bayesian statistics are about modifying degree of belief in a certain outcome given new information
- Furthermore, Frequentist models assume fixed model parameters, while Bayesian methods treat parameters as a distribution and random, reflecting our uncertainty about the prior data

> **Important**
>
> While not mutually exclusive, it's important to realize that these define probability differently, and thus generate controversy on which is 'right'

- For a more robust description of the long-running debate, see Bayarri & Berger (2004)

4 Prevalent Use of Frequentist Statistics

4.1 Examples of Frequentist Statistics

4.1.1 In Concept

- z-tests, t-tests, ANOVA - any parametric model that produces a p-value, as the p-value itself is a judgment of long-run frequency or probability of the event being "real"
- Due, to their non-parametric nature, tests like χ^2 don't mesh as well with the definition of frequentist, but the production of the p-value still means they do!

4.1.2 In Research

> **Important**
>
> Pretty much published, empirical neuropsychology article will contain use of Frequentist tests!

- To use some brief examples from recent NAP journal clubs:
  - Whitaker et al. (2024) used several t-tests and chi-square tests when determining if there were significant differences in brain cancer patients on the CVLT-C and the ChAMP.
  - Vyhnalek et al. (2015) used ANOVA, χ^2 tests, and linear regression to identify differences in Olfaction between several different MCI groups
- In both prior examples, the frequentist models were used to answer questions about comparing groups to one another and the likelihood of them being as different as they were, if the null was true.

4.1.3 In Clinic?

- Frequentist methods usually require multi-person samples for analysis, so it's use is limited at the single sample level
- However, one use is in the generation of a p-value for a binomial distribution on forced choice performance validity tests, like the VSVT (Resch et al., 2021)

4.2 Benefits of Frequentist Statistics

- As the longstanding predominant statistical method for the last 100 years (Efron, 2005), frequentist statistic enjoy wide acceptance as suitable evidence for backing claims of impact and differences
- These analyses provide concrete statements about the likelihood of a Type I error in judgment occurring - in the form of the p-value
- Practical in studying groups, and relatively simple to interpret and draw conclusions from

4.3 Drawbacks of Frequentist Statistics

- Limited usefulness in small-n sizes or single-case studies, such as when analyzing a single patient in a clinical setting
- Continuing problems in p-value misinterpretation and misuse (Lakens, 2021)

- One proposed solution is lowering of the "standard" α value (Ioannidis, 2018)

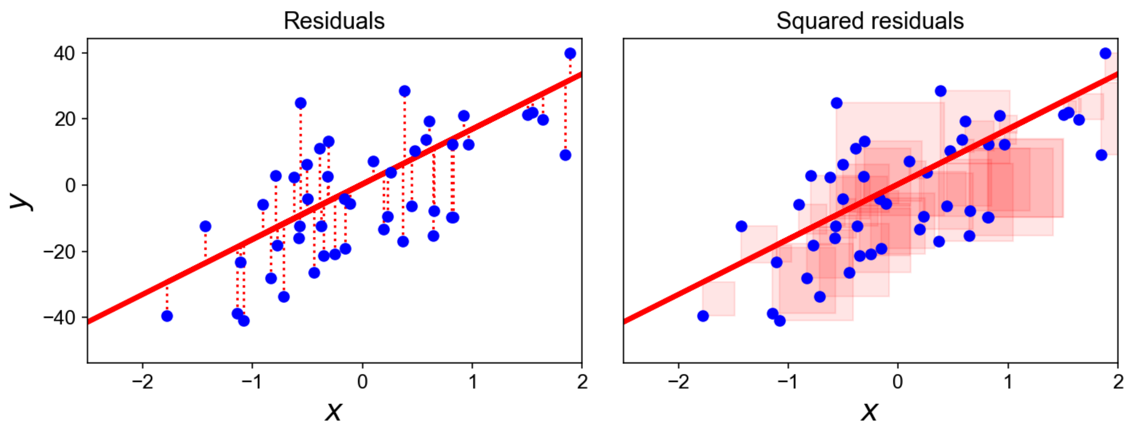
> **Important**
>
> While I am not out here to argue against the frequentist approach, this is one of the main areas where it fails to offer a practical use in clinical application!

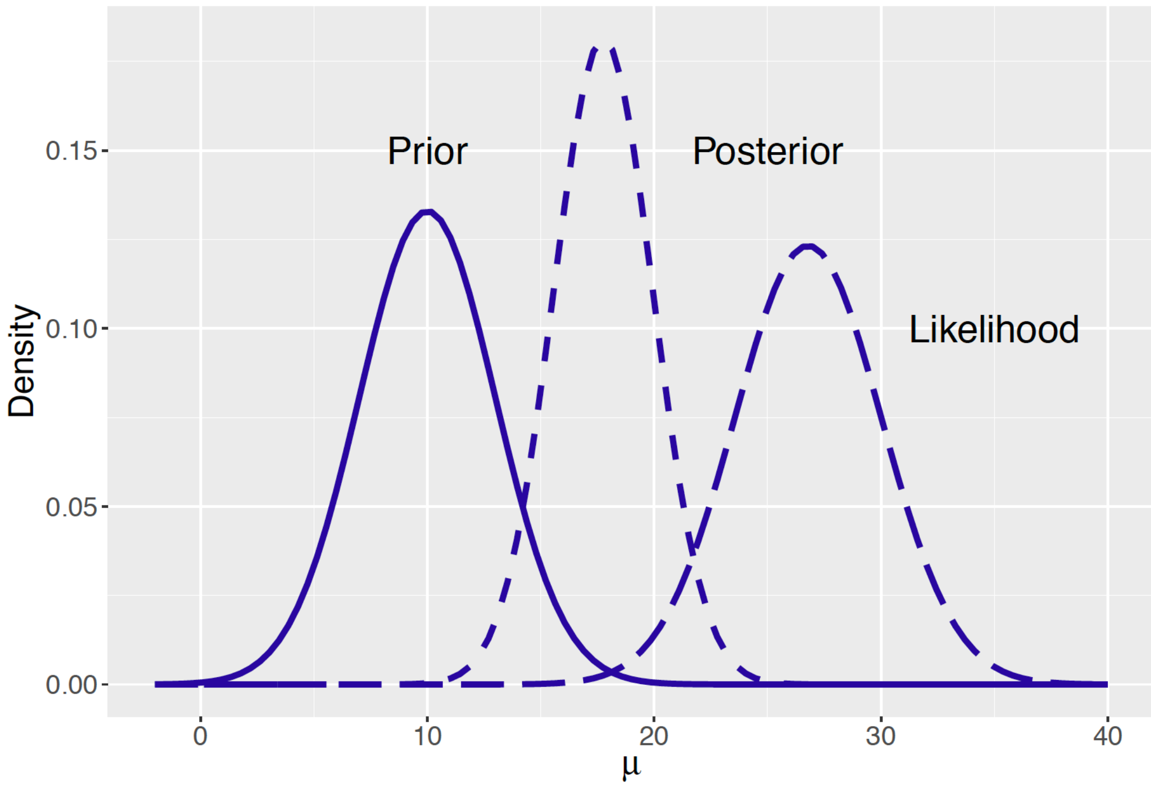
5 Applications of the Bayesian Method

5.1 Examples of Bayesian Statistics

5.1.1 In Concept

- The results of a Bayesian analysis are often presented as a set of distributions, showing the prior and posterior distributions as being impacted by the new evidence
- The most common form of Bayesian-based analysis is rooted in use of something called a Monte Carlo Markov Chain (MCMC) algorithm
  - This is the analog to something like the ordinary least squares (OLS) method or maximum likelihood method of obtaining estimates, but for Bayesian analysis





> **Important**
>
> If you see any mention of MCMC, you are likely dealing with some form of Bayesian methods!

5.1.2 In Research

- Modeling declines in cognition and/or failures in thinking as a Bayesian process with flawed evidence and prior (Parr et al., 2018)
- In same vein as the last point, can we refine or challenge existing theories by leveraging more computational methods (Schmerwitz & Kopp, 2024)
- Proposing a use for Bayesian models that can define "cutoffs" in number of tests to define as abnormal or clinically concerning (Gavett, 2015)

5.1.3 In Clinic?

- Scandola & Romano (2021) and Crawford & Garthwaite (2007): both focus on how to do robust comparison of a likely distribution of a person with a normative distribution using Bayesian methods.

  - This is similar to how we would use regular normative samples, but has the added benefit of capturing the continued uncertainty in a distribution, rather than a single point estimate with confidence intervals

- Slow but steady growth in learning and adoption, in response to broader concerns around the accuracy and stability of neuropsychological assessment (Huygelier et al., 2022)
- Emerging evidence that a well-made Bayesian analysis may provide more specific and sensitive results that align well with classification of Alzheimer's disease (Goette et al., 2023)

5.2 Benefits of Bayesian Statistics

- Still offer the ability to judge someone against a control (healthy) group, much like with regular, normative datasets
- Integrates evidence while still maintaining healthy uncertainty about the range of outcomes; remember, there is a distribution of a random parameter, not a fixed one. Avoids providing a (seemingly) authoritative point estimate
- Instead of starting at a *naive* point, like frequentist testing, Bayesian testing leverages information about where a person's distribution may lie as the prior
- Can provide predictions of performance, much like a regression model. This may offer a more algorithmic approach to evidence integration than clinical judgment alone. Data driven and algorithmic medicine has become a very popular tool to increase efficiency and accuracy (Topol, 2019)

> **Important**
>
> Data and analysis are not a replacement for the sound clinical judgement of a well-trained and licensed provider.

5.3 Drawbacks of Bayesian Statistics

- Not as widely known or supported by other scientists; less likely to be accepted as analysis in papers and presentations
- Relatively difficult to interpret and explain, due to usually more complex, computerized calculations and larger, more complicated mathematical formulas.
- Implementation for clinical practice requires well-informed understanding of Bayesian models, as selection of appropriate prior distribution is necessary
  - This is similar to a process such as norms selection, e.g., Mitrushina vs. Heaton

- Results are not as clear-cut to describe (compared to frequentist results), especially to non-scientific audiences

6 Conclusion

6.1 Further Reading & Resources

- Books
  - For a friendly and more basic introduction to the Bayesian Method → *A Student's Guide to Bayesian Statistics* (Lambert, 2018)
  - For broad coverage of statistics topics leading up to Bayesian Methods → *Statistical Rethinking, 2nd Edition* (McElreath, 2020)
  - For more focused applications of Bayesian Methods to psychology → *Bayesian Statistics for the Social Sciences, 2nd Edition* (Kaplan, 2024)
- Review the References of this presentation for scientific articles that show these methods in action!
- For more machine-learning examples, check out my 2025 NAP slides on advancements in machine learning and AI in Neuropsychology

6.2 Take-home Message

- Bayesian statistics differ greatly from frequentist statistics in the framing and questioning of probabilities
- While frequentist statistics do remain prominent, advanced methodologists are seeing more innovating ways to apply Bayesian methods to success in neuropsychological research
- Because of their unique position, Bayes-inspired methods offer a meaningful way to pursue new advances in research and clinical thinking
- While innovative, there are still many challenges in practically applying these ideas to the daily work of psychometrists
- Consider this: which perspective feels like a better way to represent probability, to you?

> **Important**
>
> It is a fallacy to insist that one method is right or wrong over the other - but they do provide different sets of benefits and drawbacks!

6.3 Contact Information

Thank you for your attention today - I welcome you to stay in touch with me to talk about this topic and other exciting neuropsychology!

- Email: Quinton.Quagliano@trinity-health.org
- Find me on LinkedIn!
- *If you wish to re-use or cite today's presentation, please follow all customary rules on attribution to myself*

Any Questions?

7 Additional Information

7.1 System and Program Information

Replicability of this file is dependent on having all necessary system dependencies. These include, but are not limited to, a TeX distribution, Pandoc, R, all listed R packages and their dependencies, and many more. Please ensure that your system is capable of knitting other Quarto documents and running R code.

Alternatively, the build environment can be replicated using the included nix flake file, which can replicate the exact TeX, Quarto, and R package versions used in this analysis. Please see the section on Reproducability Information.

This work has been version controlled with git, to improve ability to review history of files. All source code, files needed to reproduce, and .git files will be included with this document for transparency. The source code used to generate this document, and it's accompanying presentation, can be found on Github.

See Table 1 for system, R, and package information regarding the system that this document was originally rendered on.

Table 1: Technical Information

(a) System and R Information

	Specifications
platform	x86_64-pc-linux-gnu
arch	x86_64
os	linux-gnu
system	x86_64, linux-gnu
status	
major	4
minor	5.3
year	2026
month	03
day	11
svn.rev	89597
language	R
version.string	R version 4.5.3 (2026-03-11)
nickname	Reassured Reassurer

(b) Packages

Package	Version
magrittr	2.0.4
sessioninfo	1.2.3

7.2 Reproducability Information

Steps have been taken to make this document reproducible and able to be audited. Prior to attempting the steps in reproducing this work and generated PDF/presentation, please acquire/install any necessary tools and packages, as listed in Section 7.1. Because of limitations in testing time, it cannot be guaranteed that this work is entirely reproducible - please contact the author using information from the Contact Information section to get in touch if anything seems off.

7.3 References

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Key Terms

A

ANOVA A parametric family of tests used for analysis of variance among 3 or more groups on a continuous variable 8

B

Bayes Theorem A theorem that describes the Bayesian process of updating the degree of belief based upon new information 7

Bayesian statistics A framework of statistics built up the belief that probability is the degree of belief of an event 7

binomial distribution A special type of discrete numeric distribution with fail/pass and subsequent independent trials 9

F

Frequentist statistics A framework of statistics built on the belief that probability is the occurrence of something happening in an infinite number of trials 6

L

likelihood function The likelihood of the evidence given the prior information 7

N

non-parametric A description of a certain statistical test that does not rely on population parameters to inference, sometimes for power benefit. 8

P

p-value A value which describes the liklihood of a result being due to random chance, if the null hypothesis is true 7
posterior probability The degree of belief after receiving new information 7
prior probability The degree of belief prior to receiving new information 7
probability A description of how likely something is to occur, defined differently under Frequentist and Bayesian frameworks 6

T

t-tests A parametric family of tests used to compare groups or a group against a set standard when the parameters of the population are not known 8
Type I error An error in frequentist hypothesis testing due to rejecting the null hypothesis, when it is actual true 9

Z

z-tests A parametric family of tests used to compare groups or a group against a set standard when the parameters of the population are known 8